

9.C Graph Rational Functions and Identify Their Features

Rational functions contain important structural features that define their graphs. While asymptotes describe behavior near undefined values, intercepts and holes provide additional information needed to fully understand the function.

By analyzing the numerator and denominator, we can determine where a function crosses the axes, where it is undefined, and how it behaves near those points. This allows us to predict the entire graph before sketching.

"Every feature of a rational function comes from its structure."

Learning Targets

I can find x-intercepts of rational functions by setting the numerator equal to zero and solving.

I can find y-intercepts of rational functions by substituting $x = 0$ and simplifying.

I can determine all features of a rational function, including intercepts, asymptotes, holes, and behavior near asymptotes.

Introduction

In previous sections, we focused on how rational functions behave near values where they are undefined. While asymptotes describe this behavior, they are only part of the full picture.

To completely understand a rational function, we must also determine where the graph crosses the axes, where it has missing points, and how it behaves near its asymptotes. These features work together to define the structure of the graph.

Key Idea: A rational graph is determined by its structure—intercepts, asymptotes, holes, and behavior.

Connections

Prior Knowledge:

- Factoring polynomials and identifying zeros
- Asymptotes from Section 9.B
- Understanding function notation and domain restrictions

Cross-Disciplinary Connections:

- Physics: systems approaching limits or undefined conditions
- Chemistry: concentration and rate relationships involving ratios
- Computer Science: behavior near undefined inputs or limits

Real-World Applications:

- Situations where values cannot exist at certain inputs
- Modeling systems with restrictions or discontinuities
- Understanding how systems behave near critical points

History

As mathematicians expanded the study of functions, they began analyzing expressions that were undefined at certain values. These early investigations revealed that some functions behaved differently near these points, leading to the concept of discontinuities and asymptotic behavior. During the development of algebra and calculus, rational functions became important examples of functions that could not be understood by plotting points alone. Instead, their structure had to be analyzed algebraically to determine key features such as intercepts, asymptotes, and undefined values.

Today, this structural approach is essential in mathematics, science, and engineering, where understanding how a system behaves near restrictions is just as important as knowing its exact values.

Key Concepts

x-Intercepts

x-intercepts occur where the function equals zero. For rational functions, this happens when the numerator equals zero, provided the denominator is not zero at the same value.

Process:

- Set the numerator equal to zero
- Solve for x
- Check that the denominator is not zero

Example:

$$f(x) = \frac{x - 3}{x + 2} \Rightarrow x = 3$$

y-Intercepts

The y-intercept occurs where $x = 0$, as long as the function is defined at that value.

Process:

- Substitute $x = 0$
- Simplify
- Ensure the denominator is not zero

Example:

$$f(x) = \frac{x - 2}{x + 1} \Rightarrow f(0) = \frac{-2}{1} = -2$$

Holes

Holes occur when a factor appears in both the numerator and denominator and cancels. The function is undefined at that value, but it does not create a vertical asymptote.

Process:

- Factor numerator and denominator completely
- Cancel common factors
- The canceled value creates a hole

Example:

$$\frac{x-2}{(x-2)(x+1)} = \frac{1}{x+1}$$

Important: A canceled factor creates a hole, not an asymptote.

Vertical Asymptotes

Vertical asymptotes occur where the denominator equals zero after simplification.

Process:

- Factor completely
- Cancel common factors
- Set remaining denominator equal to zero

Example:

$$\frac{x-2}{(x-2)(x+1)} = \frac{1}{x+1} \Rightarrow x = -1$$

Horizontal and Slant Asymptotes

End behavior is determined by comparing the degrees of the numerator and denominator.

Case	Result
Degree (top) < Degree (bottom)	$y = 0$
Degrees equal	Ratio of leading coefficients
Top degree is one higher	Slant asymptote

Example:

$$\frac{x^3 - 1}{x^2 - 1} = x + \frac{x - 1}{x^2 - 1} \Rightarrow y = x$$

Behavior Near Asymptotes

Near vertical asymptotes, the function increases or decreases without bound. The direction depends on the sign of the function near the asymptote.

Key Idea: Determine whether the function approaches $+\infty$ or $-\infty$ on each side.

Full Structural Analysis

To completely understand a rational function, all features must be identified before graphing.

Complete Process:

1. Factor numerator and denominator completely
2. Simplify to identify holes
3. Find x-intercepts
4. Find y-intercept
5. Find vertical asymptotes
6. Determine horizontal or slant asymptotes
7. Analyze behavior near asymptotes

Vocabulary

Rational Function:

A function written as a ratio of two polynomials.

x-Intercept:

A point where the graph crosses the x-axis, found by setting the numerator equal to zero.

y-Intercept:

A point where the graph crosses the y-axis, found by substituting $x = 0$.

Hole:

A point where the function is undefined due to a canceled factor, but does not create an asymptote.

Vertical Asymptote:

A vertical line where the function is undefined after simplification and the graph approaches but does not cross.

Horizontal Asymptote:

A horizontal line that describes the end behavior of a function as x becomes large.

Slant Asymptote:

A diagonal line that describes end behavior when the degree of the numerator is one greater than the denominator.

Discontinuity:

A point where a function is not defined, including holes and asymptotes.

Strategies

Complete Feature Analysis:**1. Factor completely.**

Factor both the numerator and denominator to reveal structure.

2. Simplify first.

Cancel any common factors to identify holes.

3. Identify holes.

Any canceled factor creates a hole, not an asymptote.

4. Find x-intercepts.

Set the simplified numerator equal to zero.

5. Find the y-intercept.

Substitute $x = 0$, if defined.

6. Find vertical asymptotes.

Set the remaining denominator equal to zero.

7. Determine end behavior.

Use degrees to find horizontal or slant asymptotes.

8. Analyze behavior near asymptotes.

Determine whether the function approaches $+\infty$ or $-\infty$.

9. Use structure before graphing.

All key features should be identified algebraically first.

| Examples

Example 1 — Intercepts

Find all intercepts.

$$f(x) = \frac{x - 4}{x + 1}$$

Example 2 — Hole

Simplify and identify any holes.

$$\frac{x - 2}{(x - 2)(x + 3)}$$

Example 3 — Full Feature Analysis

Identify all features.

$$f(x) = \frac{x^2 - 9}{x^2 - x - 6}$$

Example 4 — Cubic Structure + Slant

Identify all features.

$$f(x) = \frac{x^3 - 1}{x^2 - 1}$$

Example 5 — Factor, Simplify, Then Analyze

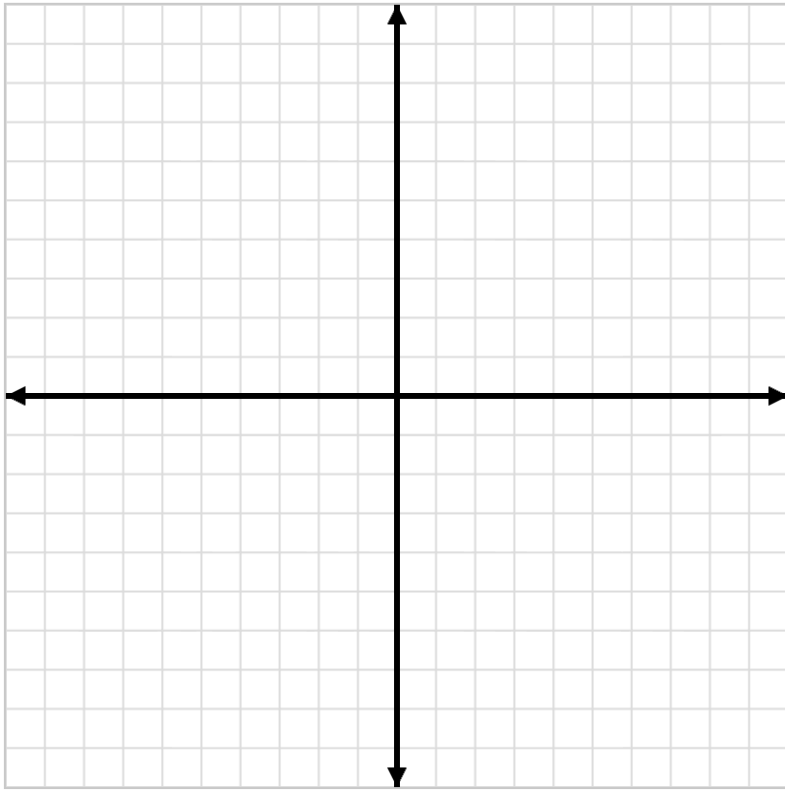
Factor, simplify, and identify all features.

$$f(x) = \frac{x^3 - 4x^2 + 3x}{x^2 - 3x + 2}$$

Example 6 — Graph from Features

Identify all features, then sketch the graph.

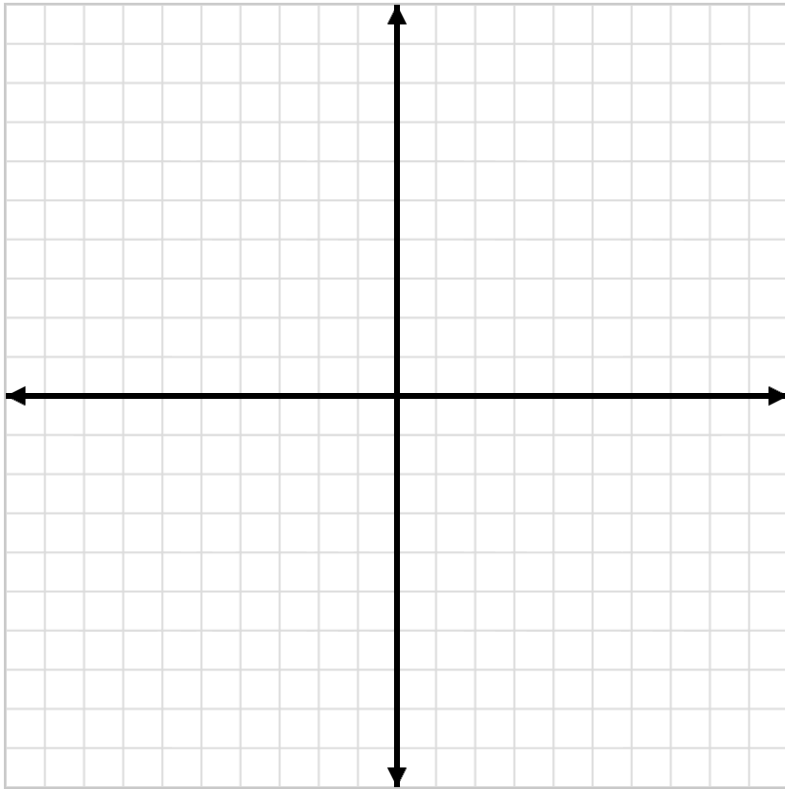
$$f(x) = \frac{x(x-2)}{(x-2)(x+3)}$$



Example 7 — Full Graph (Advanced)

Identify all features and sketch the graph.

$$f(x) = \frac{x^2 - 4}{x^2 - 5x + 6}$$



Common Mistakes

- **Not simplifying first:**

Failing to factor and cancel before analyzing leads to incorrect asymptotes and missed holes.

- **Confusing holes with vertical asymptotes:**

If a factor cancels, it creates a hole—not a vertical asymptote.

- **Including undefined values as intercepts:**

A value that makes the denominator zero cannot be an intercept.

- **Incorrect x-intercepts:**

x-intercepts come only from the numerator and must be checked against the denominator.

- **Forgetting to check $x = 0$:**

The y-intercept exists only if the function is defined at $x = 0$.

- **Skipping behavior near asymptotes:**

Knowing the asymptote is not enough—you must determine how the graph approaches it.

- **Trying to graph too early:**

All features should be determined algebraically before sketching.

Summary

- x-intercepts are found by setting the numerator equal to zero.
- y-intercepts are found by substituting $x = 0$, if the function is defined.
- Holes occur when factors cancel between numerator and denominator.
- Vertical asymptotes come from remaining denominator factors after simplification.
- Horizontal and slant asymptotes describe end behavior.
- Behavior near asymptotes determines how the graph approaches these lines.
- Complete structure must be identified before graphing.

| Practice Set

Level 1 — Easy

1. Find the x-intercept.

$$f(x) = \frac{x - 2}{x + 1}$$

2. Find the x-intercepts.

$$f(x) = \frac{x^2 - 9}{x + 2}$$

3. Find the y-intercept.

$$f(x) = \frac{2x + 3}{x - 1}$$

4. Find the y-intercept.

$$f(x) = \frac{x^2 - 1}{x + 3}$$

5. Determine if an x-intercept exists.

$$f(x) = \frac{1}{x + 4}$$

Level 2 — Medium

6. Find all intercepts and vertical asymptotes.

$$f(x) = \frac{x - 3}{x - 2}$$

7. Find all intercepts and vertical asymptotes.

$$f(x) = \frac{x + 1}{x^2 - 4}$$

8. Find all asymptotes.

$$f(x) = \frac{2x^2}{x^2 + 1}$$

9. Find all asymptotes.

$$f(x) = \frac{x^2 + 1}{x - 3}$$

10. Identify all features.

$$f(x) = \frac{x - 2}{x + 3}$$

Level 3 — Hard

11. Simplify and identify all features.

$$\frac{x - 3}{(x - 3)(x + 2)}$$

12. Simplify and identify all features.

$$\frac{x + 4}{(x + 4)(x - 1)}$$

13. Find intercepts, asymptotes, and holes.

$$f(x) = \frac{x(x-2)}{(x-2)(x+1)}$$

14. Find intercepts, asymptotes, and holes.

$$f(x) = \frac{(x+3)(x-1)}{(x+3)(x-4)}$$

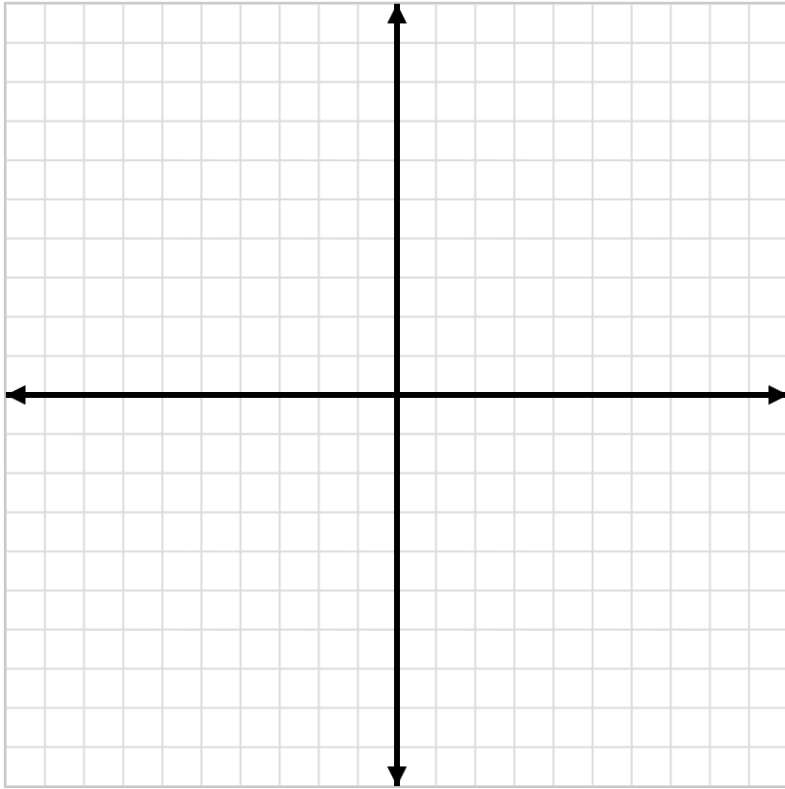
15. Describe behavior near the asymptote.

$$f(x) = \frac{2}{x-3}$$

Level 4 — Challenging

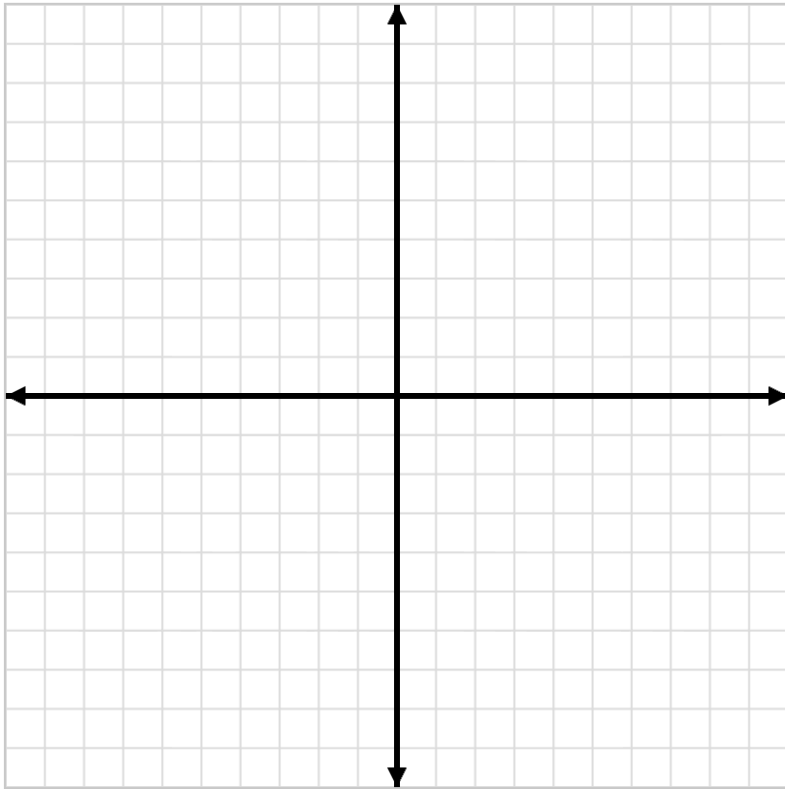
16. Factor completely, identify all features, and sketch the graph.

$$f(x) = \frac{x^3 - 6x^2 + 11x - 6}{x^2 - 5x + 6}$$



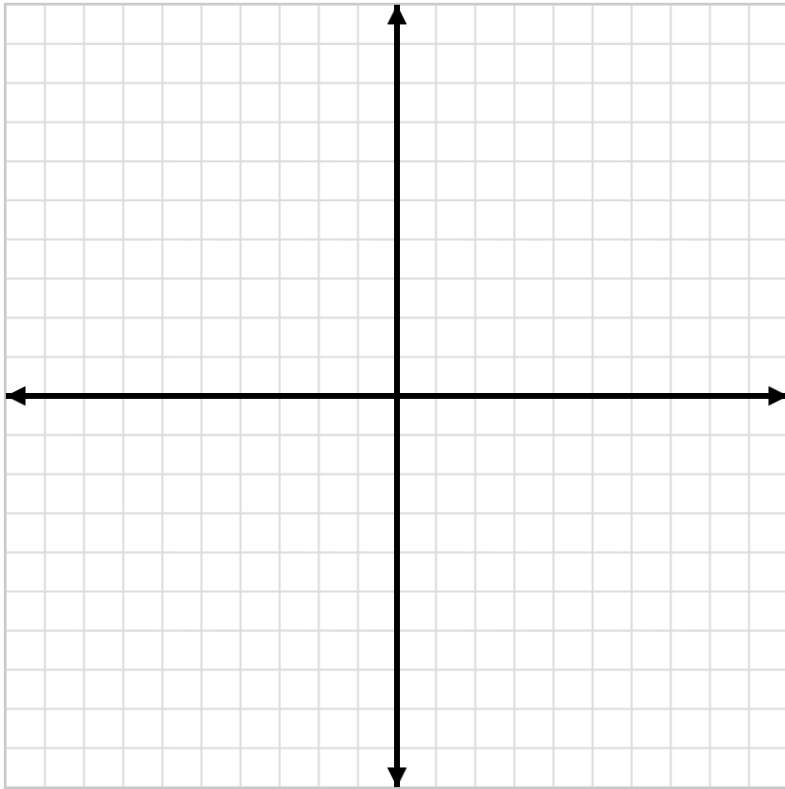
17. Factor completely, identify all features, and sketch the graph.

$$f(x) = \frac{x^3 - 4x^2 - x + 4}{x^2 - 4}$$



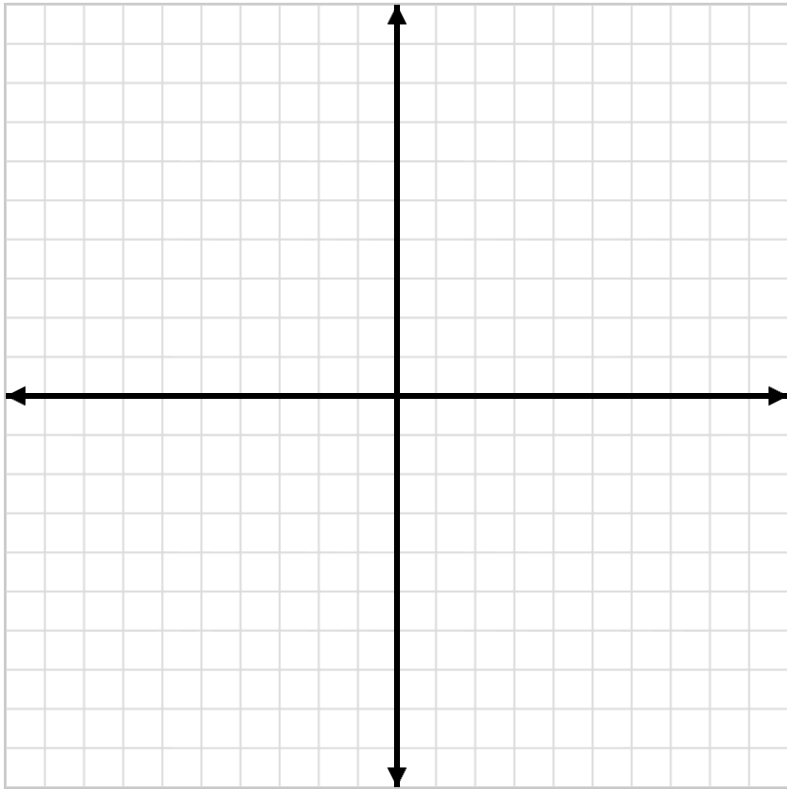
18. Factor, simplify, identify all features, and sketch the graph.

$$\frac{x^3 - 3x^2 + 2x}{x^2 - 3x + 2}$$



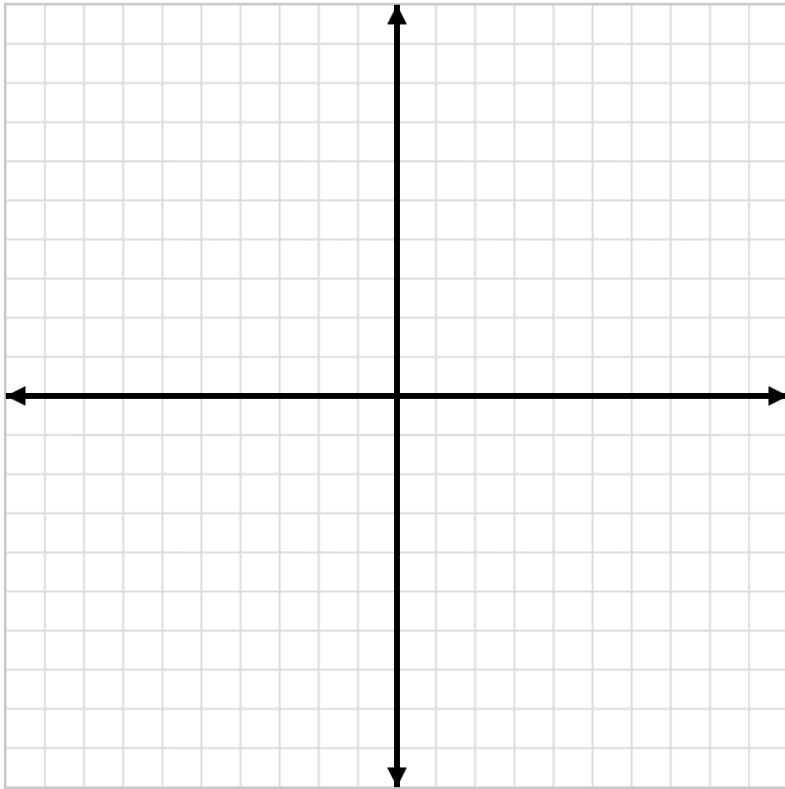
19. Factor, simplify, identify all features, and sketch the graph.

$$\frac{x^3 - 9x}{x^2 - 3x}$$



20. Identify all features and sketch the graph. Clearly show behavior near asymptotes.

$$f(x) = \frac{x^2 - 4}{x^2 - 5x + 6}$$



Level 5 — Justification / Conceptual

21. Explain why x-intercepts come from the numerator.

22. Explain why holes occur instead of asymptotes.

23. Explain how asymptotes affect the shape of a graph.

24. Explain why some rational functions have no x-intercepts.

25. Describe a complete process for analyzing a rational function before graphing.

Practice Answer Key

Level 1 — Easy

1. $x=2$
2. $x=\pm 3$
3. $y=-3$
4. $y=-\frac{1}{3}$
5. none

Level 2 — Medium

6. x-int: 3, VA: $x=2$
7. x-int: -1, VA: $x=\pm 2$
8. HA: $y=2$
9. VA: $x=3$, SA: $y=x+3$
10. x-int: 2, VA: -3, HA: 1

Level 3 — Hard

11. hole at $x=3$, VA: -2, HA: 0
12. hole at $x=-4$, VA: 1, HA: 0
13. hole at $x=2$, VA: -1, x-int: 0, HA: 0
14. hole at $x=-3$, VA: 4, x-int: 1, HA: 0
15. approaches $\pm\infty$ near $x=3$

Level 4 — Challenging

16. holes at $x=1,2,3 \rightarrow$ simplifies to linear; graph line with holes
17. factor $\rightarrow$ cancellation + VA at $x=\pm 2$; slant asymptote present
18. simplifies; hole at $x=1,2$; reduced linear behavior
19. hole at $x=0$, VA at $x=3$; remaining linear structure
20. VA: $x=2,3$; HA: $y=1$; x-int: ± 2

Level 5 — Justification

21–25. Answers vary (conceptual)

Detailed Example Solutions

Example 1 — Intercepts

$$f(x) = \frac{x - 4}{x + 1}$$

Set numerator equal to zero:

$$x - 4 = 0 \Rightarrow x = 4$$

x-intercept: (4,0)

Find y-intercept:

$$f(0) = \frac{-4}{1} = -4$$

y-intercept: (0,-4)

Example 2 — Hole

$$\frac{x - 2}{(x - 2)(x + 3)}$$

Factor and cancel:

$$= \frac{1}{x + 3}$$

Hole at $x = 2$

Find y-value:

$$f(2) = \frac{1}{5}$$

Hole: $(2, \frac{1}{5})$

Example 3 — Full Feature Analysis

$$f(x) = \frac{x^2 - 9}{x^2 - x - 6}$$

Factor:

$$\frac{(x - 3)(x + 3)}{(x - 3)(x + 2)}$$

Cancel:

$$= \frac{x + 3}{x + 2}$$

Hole at $x = 3$

Vertical asymptote:

$$x = -2$$

x-intercept:

$$x = -3$$

Horizontal asymptote:

$$y = 1$$

Example 4 — Cubic Structure + Slant

$$f(x) = \frac{x^3 - 1}{x^2 - 1}$$

Factor:

$$\frac{(x - 1)(x^2 + x + 1)}{(x - 1)(x + 1)}$$

Cancel:

$$= \frac{x^2 + x + 1}{x + 1}$$

Hole at $x = 1$

Divide:

$$= x + \frac{1}{x + 1}$$

Slant asymptote: $y=x$

Vertical asymptote: $x=-1$

Example 5 — Factor, Simplify, Then Analyze

$$f(x) = \frac{x^3 - 4x^2 + 3x}{x^2 - 3x + 2}$$

Factor:

$$\frac{x(x-1)(x-3)}{(x-1)(x-2)}$$

Cancel:

$$= \frac{x(x-3)}{x-2}$$

Hole at $x = 1$

Vertical asymptote:

$$x = 2$$

Example 6 — Graph from Features

$$f(x) = \frac{x(x-2)}{(x-2)(x+3)}$$

Simplify:

$$= \frac{x}{x+3}$$

Hole at $x = 2$

Vertical asymptote:

$$x = -3$$

x-intercept:

$$x = 0$$

Horizontal asymptote:

$$y = 1$$

Example 7 — Full Graph

$$f(x) = \frac{x^2 - 4}{x^2 - 5x + 6}$$

Factor:

$$\frac{(x - 2)(x + 2)}{(x - 2)(x - 3)}$$

Cancel:

$$= \frac{x + 2}{x - 3}$$

Hole at $x = 2$

Vertical asymptote:

$$x = 3$$

x-intercept:

$$x = -2$$

Horizontal asymptote:

$$y = 1$$